Bags, Vectors & Transformers

The Hugging Face Hub · Day 3, Part 3

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Hugging Face: The Practical Gateway

You do not train these from scratch or hunt down files. **Hugging Face** is a hub of **thousands of pre-trained models**, plus the `transformers` library to use them.

The Model Hub 🔗

- Search by **task** or **language**
- Ready-made & fine-tuned models
- Model cards document training & limits
- Mostly free to download

The `transformers` library

- Load any model in **a few lines**
- The `pipeline` helper for quick tasks
- Full control for fine-tuning
- Works seamlessly in Colab

Sentiment in Three Lines

The `pipeline` helper hides all the complexity. Here is a working off-the-shelf sentiment classifier:

```
from transformers import pipeline

clf = pipeline("sentiment-analysis")

clf("This policy is a disaster.")

# [{'label': 'NEGATIVE', 'score': 0.99}]
```

That is a real, state-of-the-art model running on your text. Swapping `"sentiment-analysis"` for another task, or naming a specific model (say a Dutch one), is all it takes to change what it does.

Reading a Model Card

Every model on the Hub has a **model card**. It is the methods section of the model: you wouldn't use a measurement instrument without knowing how it was built.

Questions to ask

- Trained on **what**? Data, language, domain, time period
- Which **task**, which **labels**, numbered how?
- **Evaluated** how, and on which data?
- What does it say about **limitations** and **bias**?
- **License**: research? commercial?
- Who made it, and is there a **paper**?

Red flags

- No training data mentioned
- No evaluation, or no word on which data
- "More information needed" everywhere
- No license
- Labels called `LABEL_0`, `LABEL_1`, unexplained

*Some models are **gated**: you log in and accept the terms before you can download them.*

Popular Isn't the Same as Good

The Hub sorts models by **downloads** and **likes**. That tells you what's popular, not what fits your data.

Remember "Sentiment in Three Lines"? Without a model name, `pipeline("sentiment-analysis")` quietly loads a default: DistilBERT fine-tuned on **SST-2, sentences from movie reviews**. It gets downloaded a lot, precisely because it's the default. Is "This policy is a disaster." a movie review?

How to choose instead

1. Filter the Hub by **task** and **language**.
2. Read the **card** of a handful of candidates.
3. Try them on **100 to 200 of your own** hand-labeled texts.
4. Pick on that score. Then name the model explicitly in your code and your paper.

Datasets Live on the Hub, Too

Besides models, the Hub hosts **datasets**. The `datasets` library loads them in one line, with their training, validation and test splits.

```
from datasets import load_dataset

ds = load_dataset("climatebert/environmental_claims")

# train: 2,117 rows · validation: 265 · test: 265
# columns: text, label (no / yes)
```

Same questions as for models

Who labeled the data, how, and how well did they agree? Under which license? Read the **dataset card**.

Original or a copy?

Anyone can upload. Some older datasets no longer load with current libraries, and community copies fill the gap. Check how a copy was made.

Where Does Your Data Go?

Download and run

What we do today. The **model comes to you**: its files are downloaded, and your texts are processed wherever the code runs.

In Colab, that's a Google machine. On your laptop or a university server, the texts stay there.

Hosted inference

APIs, hosted endpoints, chat interfaces. Your **texts go to the model**: they are sent to someone else's servers, under their terms.

Convenient, but you give up control over where the data ends up.

Interviews, health data, anything personal: check your data management plan and consent forms before you upload texts anywhere, **Colab included**. Open models you can run on approved hardware are a big advantage here (more on Thursday).

Hub Scavenger Hunt

IN PAIRS Take one card, find a match on `huggingface.co`, and answer the four questions.

1 A **sentiment** model for **Dutch** text

2 An **emotion** classifier with more than two emotions

3 A model that detects **hate speech** or toxic language

4 A **named-entity** model for **German**

5 A model that codes text into **policy topics**

6 A labeled **dataset** of news or political texts

For your find: (a) What was it trained on? **(b)** Which license? **(c)** Reported performance, on which data? **(d)** One thing the card doesn't tell you.

Then try it on three sentences of your own with `pipeline(...)`. Report back in one minute: would you use it for research?

The Hub: Takeaways

- ✓ The Hub has **models** and **datasets**; `transformers` and `datasets` load them in a few lines
- ✓ **Read the card**: training data, labels, evaluation, limitations, license
- ✓ Downloads measure popularity, not fit. **Validate** on 100 to 200 of your own texts
- ✓ Name the model explicitly, in your code and in your paper
- ✓ Know **where your data goes**: download and run, or hosted inference

Next: your turn, on three datasets from the Hub.